



Climate Change and Global Food Security

Interactive dashboard exploring climate change and global food security (2000–2024).

Data sources

- **NASA GISS** - *Climate data*
- **EM-DAT** - *Natural disasters*
- **FAOSTAT** - *Agricultural production, trade and food prices*

Author

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Power BI Business Intelligence Project

Climate Change Overview (2000–2024)

Understanding how climate change evolved before analyzing its impact on global food trade

Temperature Anomaly

0.77 °C

Disaster Events

8K

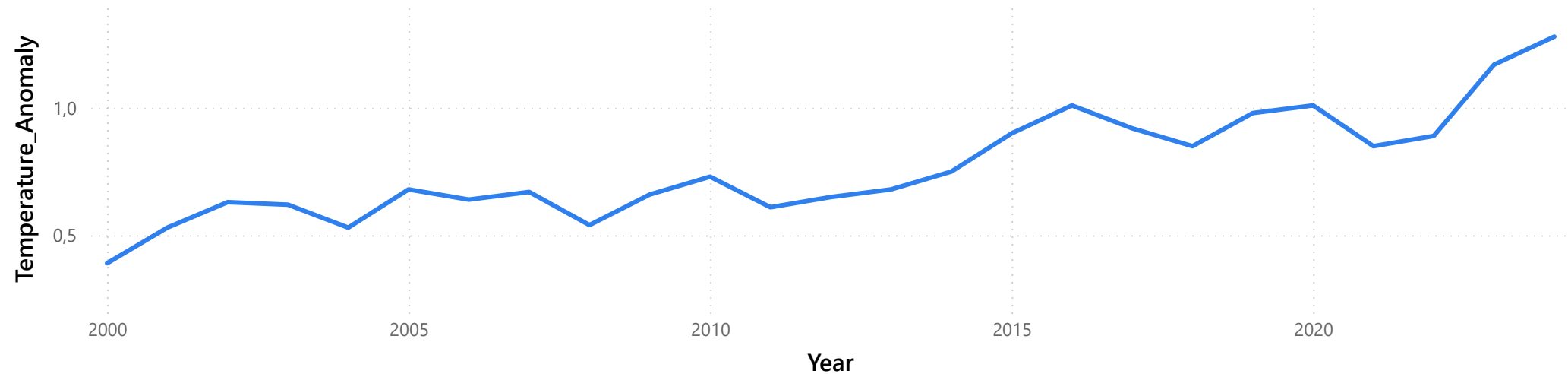
Total Economic Damage

\$3,0bn

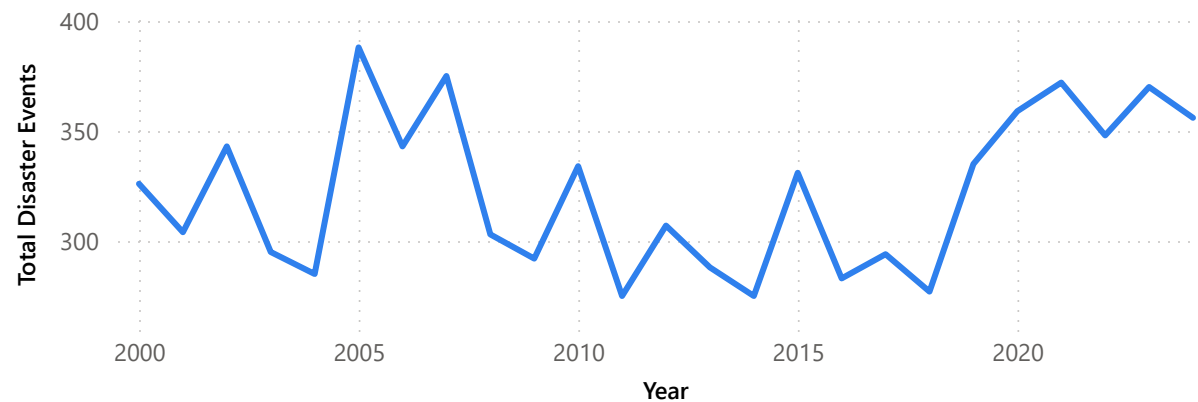
People Affected by Disaster Events

4,57bn

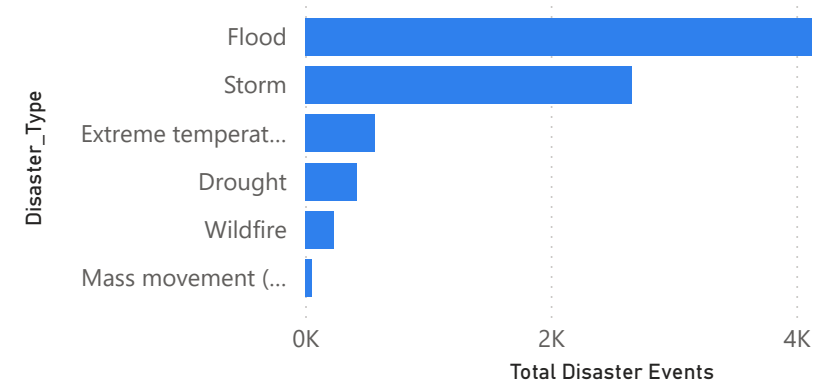
Global Temperature Anomaly Trend



Disaster Events Over Time



Top Disaster Types

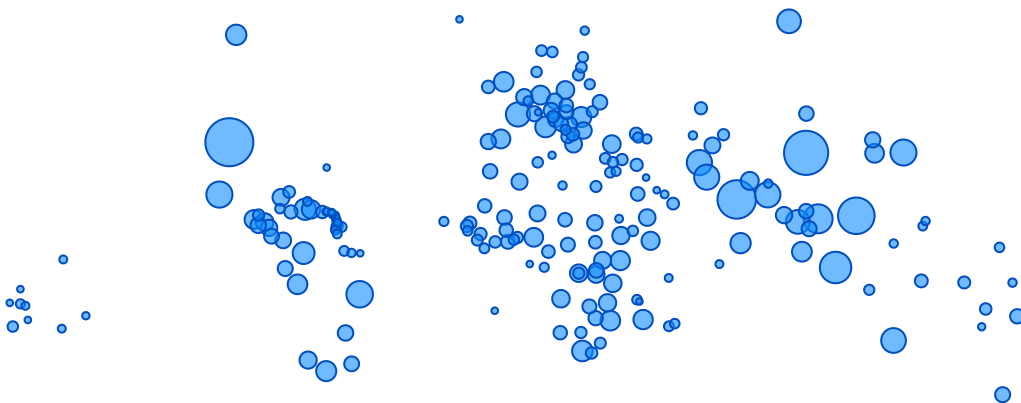


Natural Disaster Exposure

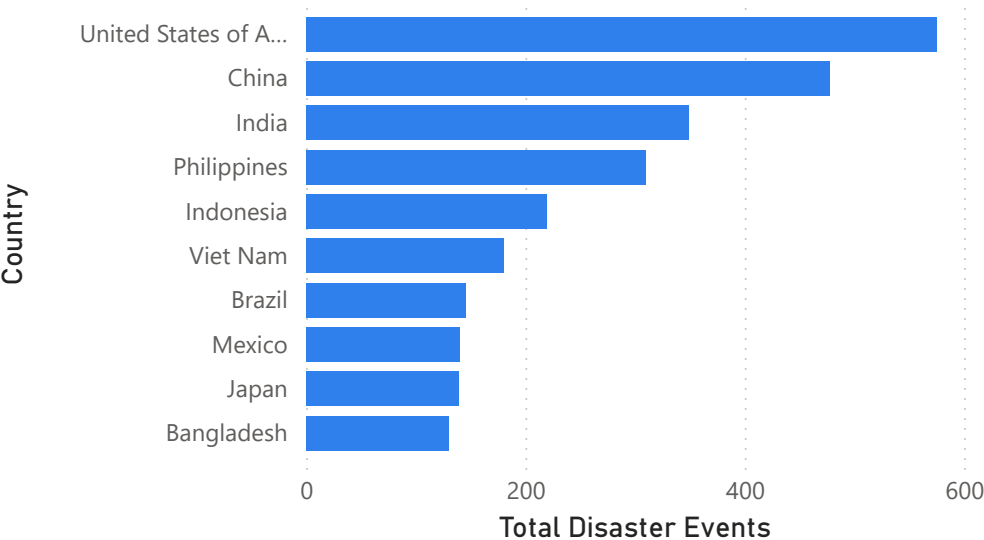
Identifying where disaster events and human impacts are concentrated across countries and disaster types



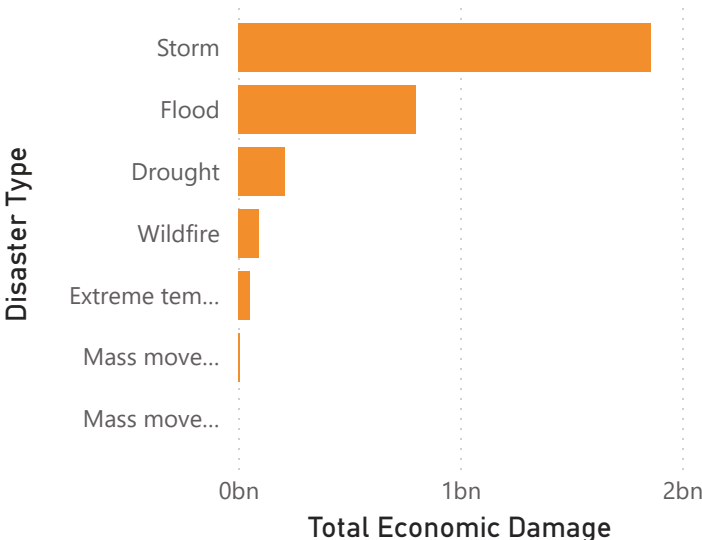
Disaster Events by Country



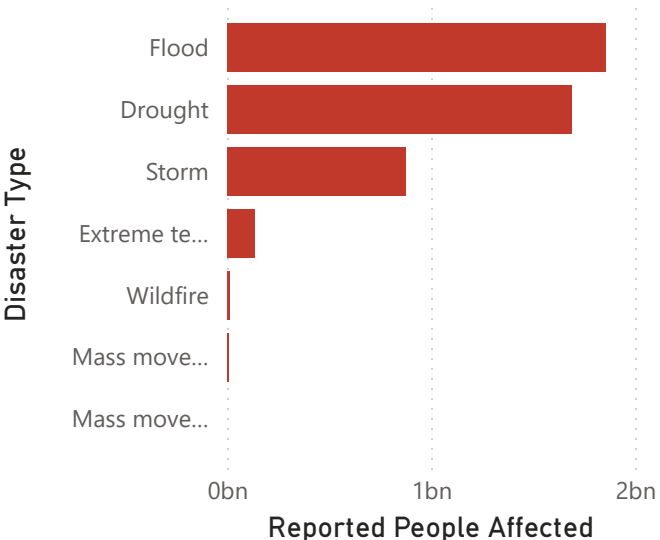
Top 10 Countries by Disaster Events



Economic Damage by Disaster Type



Reported People Affected by Disaster Type



Agricultural Production Trends

Analyzing production patterns of key food commodities to identify potential climate-related vulnerabilities

Commodity

All

Total Production

727,7bn

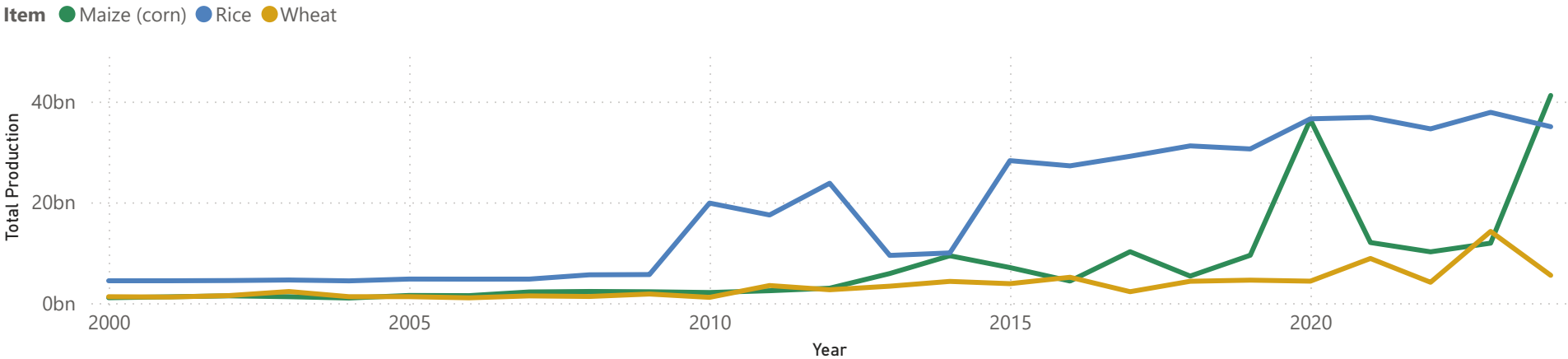
Average Annual Production

70,45M

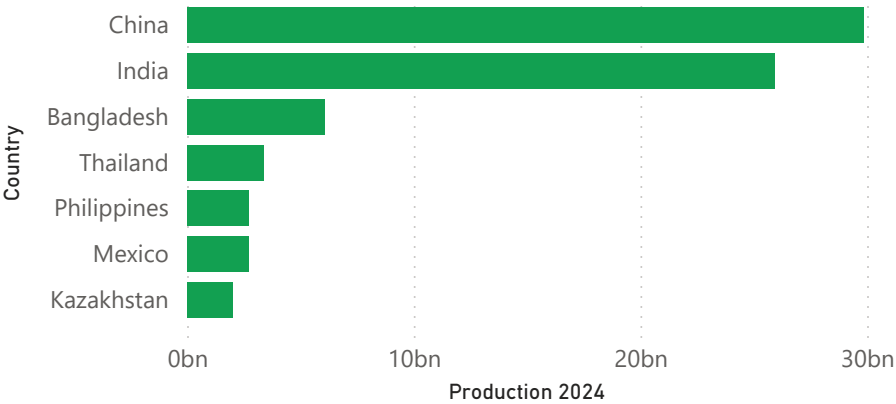
Analyzed Commodities

3

Global Production Trend by Commodity



Top Producer Countries (2024)



Total Production by Commodity



Global Food Trade Dynamics

Exploring how international trade flows of key food commodities evolved over time

Commodity

All

Total Trade Value

3,65bn

Import Value

1,98bn

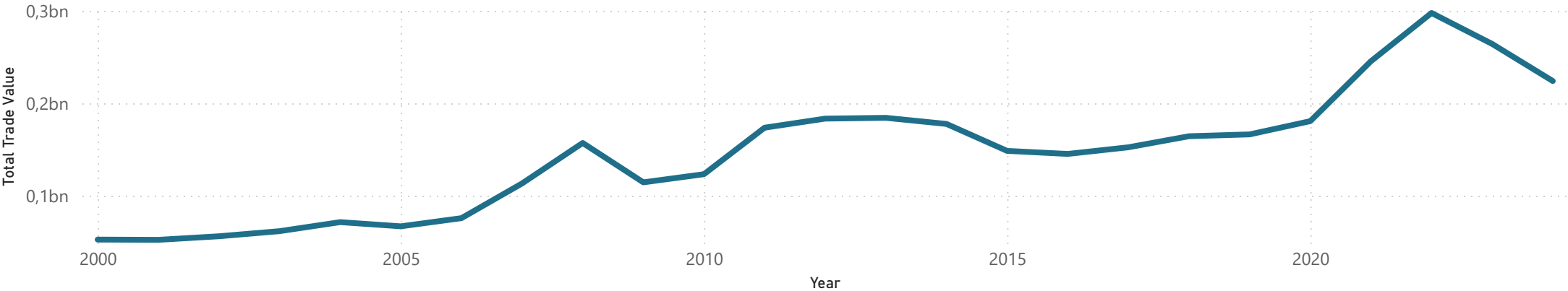
Export Value

1,67bn

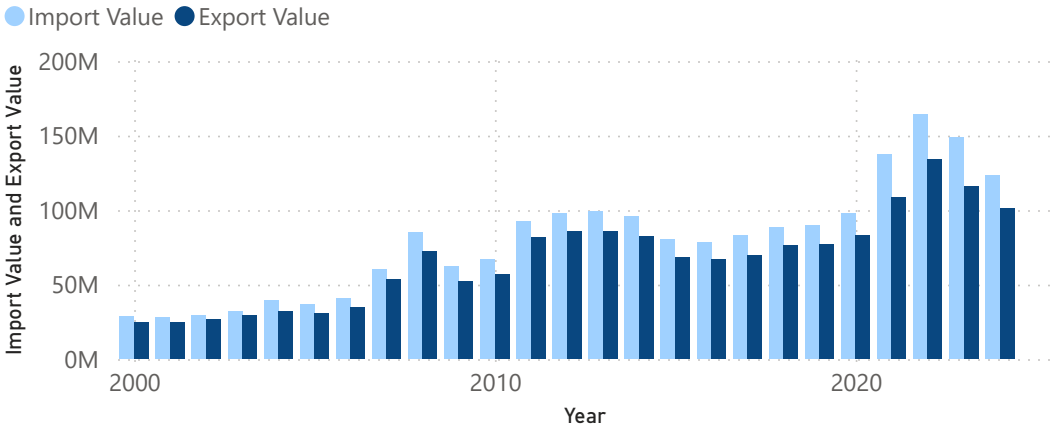
Trade Balance

-312M

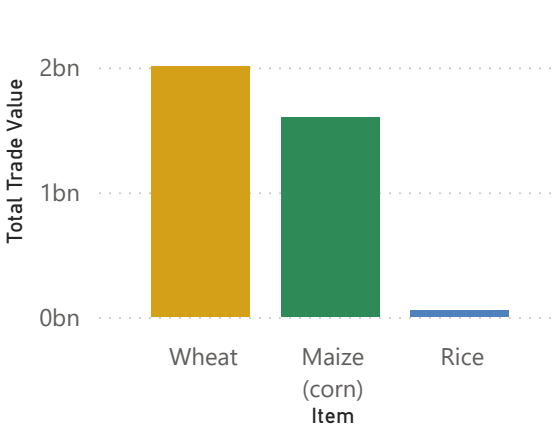
Global Food Trade Trend



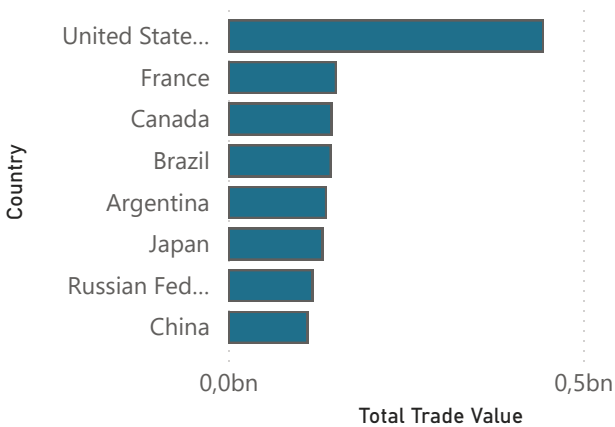
Import vs Export Over Time



Total Trade Value by Commodity



Top Trading Countries



Food Price Evolution

Understanding long-term trends in global food and cereals price indices

Average Food Price Index

96,71

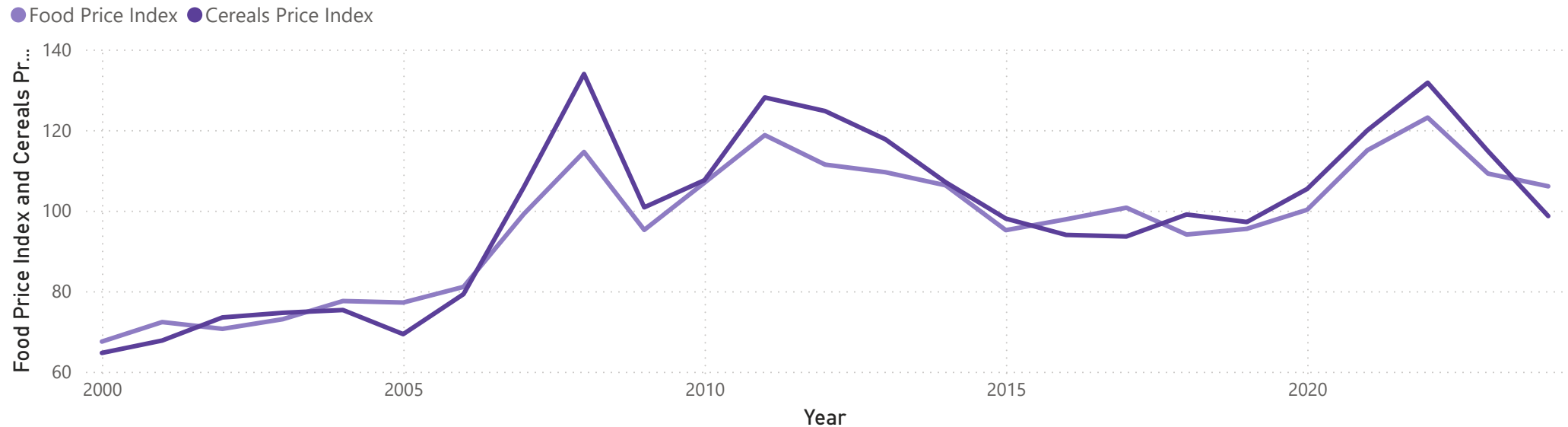
Average Cereals Price Index

99,29

Years Analyzed

25

Food and Cereals Price Trends



Key Insights

- Both indices reached major peaks around **2008** and **2022**.
- The **Cereals Price Index** showed greater volatility than the overall **Food Price Index**.
- Following the 2022 peak, both indices declined, suggesting a partial stabilization of global food markets.

Key Findings and Conclusions

Summary of the main insights derived from the analysis of climate change, natural disasters, agricultural production, trade, and food prices

Main Findings

- **Global temperatures increased steadily** between 2000 and 2024.
- **Floods and storms** were the most frequent and impactful disaster types.
- **Agricultural production continued to grow**, although trends varied across commodities.
- **International food trade expanded**, with wheat accounting for the largest trade value.
- **Food and cereals price indices experienced significant peaks** around 2008 and 2022 before stabilizing.

Overall Conclusion

The analysis suggests that climate change is associated with increasing environmental risks that may affect agricultural systems and global food markets.

Despite continuous growth in production and international trade, food price volatility highlights the importance of resilient agricultural and trade policies to support long-term food security.